

Alexandre Benoit

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Summary

ML researcher and engineer with 2+ years of research experience specialising in building, training, and debugging complex ML systems at scale. 8 peer-reviewed publications across the University of Cambridge, UCL, and Bath spanning low-precision deep learning on accelerators, graph neural networks, semi-supervised learning, multi-agent reinforcement learning, and model interpretability. Hands-on experience with GPU cluster training, mixed-precision training (BF16/FP16/FP32), large-scale synthetic data pipelines, and systematic model benchmarking with Weights & Biases. Open-source contributor for Stripe and Tempo on the Machine Payments Protocol for AI agents (npm). Strong software engineering foundation in Python, PyTorch, C/C++, and TypeScript, with production deployment experience using Docker, CI/CD, and Azure.

Research Experience

University of Cambridge, Department of Engineering, Machine Learning Research Assistant – Lab of Prof. José Miguel Hernández-Lobato Cambridge, UK
May 2025 – Sept 2025

Developed low-precision execution policies for the MACE SO(3)-equivariant graph neural network used in molecular dynamics force fields. Profiled MACE end-to-end and per-block, compared the e3nn and NVIDIA cuEquivariance backends, and assessed FP64/FP32/ BF16/ FP16 settings (with FP32 accumulation) for inference, NVT/NPT water simulations, and training runs. Demonstrated that cuEquivariance reduces inference latency by $\sim 3\times$, with an additional $\sim 4\times$ speedup from casting linear layers to BF16/FP16 within an FP32 model, while preserving energies and thermodynamic observables within run-to-run variability. Work published as arXiv preprint: *Speeding Up MACE: Low-Precision Tricks for Equivariant Force Fields* (arXiv:2510.23621).

University College London (UCL), Machine Learning Research Assistant – e-Motion Lab, Dr. Pedram Asef London, UK
2024 – Present

Led empirical ML research on energy systems with publications in *Energy and AI*. Designed a novel semi-supervised deep kernel learning framework for battery state-of-health estimation (Encoder-T + GRU + Gaussian Process Regression with custom composite LIN kernel), establishing a new state of the art on the NASA randomised battery usage dataset (24% MAPE improvement). Co-developed an AI-driven hybrid method for flicker estimation in power systems. Trained models on GPU clusters (RTX 3080 Ti) with Bayesian hyperparameter optimisation via Weights & Biases.

University of Bath, Machine Learning Engineer – Research Assistant Bath, UK
Jun 2023 – 2024

Conducted ML research on energy management and autonomous systems, resulting in three peer-reviewed publications: a survey of OR-Tools and ML for global path planning in autonomous vehicles (*Advanced Intelligent Systems*, 2024); multi-agent reinforcement learning for scheduling commercial EV charging stations (UPEC 2024); and an AI-driven hybrid method for flicker estimation in power systems (SEST 2024). Built end-to-end ML pipelines for time-series prediction, RL environment design, and deep learning model evaluation.

Industry Experience

Savanta, ML/AI Engineer & Solution Architect London, UK
Jan 2026 – Present

Architect and build Virtual Personas, Synthetic Personas and ACMD (Automatic Clustering of Market Data), production AI agentic orchestration and unsupervised machine learning platforms for market research. Built an unsupervised clustering market-segmentation pipeline replacing 1-3 months of work in hours, a synthetic-responder generation engine, and a society-simulation system, all driven by LLM agents. Own the full stack across six microservices (FastAPI, React, TypeScript, PostgreSQL, MongoDB, Redis) on Azure, with agent pipelines on LangChain, LangGraph, and Langfuse routing across OpenAI and Anthropic models. Built evaluation harnesses validating outputs against ground-truth distributions, ran 100+ prompt-fidelity benchmarks, and led a full security audit. **The products I built have generated just under £1M in client revenue in their first 5 months.** Build interactive dashboards and present findings directly to non-technical and C-suite stakeholders at enterprise clients (JP Morgan, Chase, Brompton, BPP, LVMH); coordinate delivery across a small cross-functional team.

Team Bath Racing Electric, Software Engineer – Autonomous Driving Bath, UK
Mar 2023 – Jun 2023

Developed and integrated software components for the autonomous driving stack in C++ and Python, including Model Predictive Control (MPC) algorithms, MCP integration of AI computer-vision tools for the perception pipeline, and API integration with the Dynamic Driving Task system for real-time vehicle control.

Jaguar TCS Formula E, Software Engineer Grove, Oxfordshire, UK
Sep 2021 – May 2022

Built and maintained large-scale real-time driving simulation environments in C++ and MATLAB for Formula E race strategy R&D. Developed telemetry-driven feedback systems, contributed to torque vectoring control logic, and extended the simulation toolchain with a custom web interface for engineer interaction.

Education

M.Phil, University of Cambridge, Clare College, Machine Learning and Machine Intelligence (MLMI) Cambridge, UK
Sept 2024 – Sept

- GPA: A-
- Thesis: Speeding Up MACE: Low-Precision Tricks for Equivariant Force Fields.

M.Eng, University of Bath, Integrated Mechanical and Electrical Engineering

Bath, UK
Sept 2019 – June
2024

- GPA: **First Class Honours (73%)** – top 2 of cohort – GPA 4.0/4.0
- Theses: Semi-Supervised Deep Kernel for Battery State of Health Estimation; Autonomous Mining Path Planning via Machine Learning.

Selected Projects

MACE Low-Precision Acceleration – Cambridge MPhil Thesis

2025

Demonstrated that MACE force fields can be trained and run **faster and at lower precision without sacrificing physical fidelity**. Profiled computational bottlenecks per block, compared e3nn vs NVIDIA cuEquivariance backends, and developed a practical execution policy: cuEquivariance with FP32 by default plus BF16/FP16 on linear layers (FP32 accumulations) yields up to $\sim 12\times$ combined speedup on inference, with energies and thermodynamic observables in NVT/NPT water simulations remaining within run-to-run variability. [\[Code\]](#) [\[Paper\]](#)

Semi-Supervised Deep Kernel Learning Framework – New State of the Art on NASA Battery Dataset

2025

Designed a custom Encoder-T + GRU architecture coupled with Gaussian Process Regression and a novel composite LIN kernel capturing both long-term linear capacity degradation and short-term regeneration effects, trained under a semi-supervised regime using only 15% labelled data. Systematically benchmarked 12 upstream architectures under identical conditions (15–30 runs each, Bayesian HPO via Weights & Biases) on GPU clusters. Established new SOTA on the NASA randomised battery dataset, surpassing the previous benchmark by over 24% in MAPE. Used SHAP, saliency maps, and GRU gate-weight dynamics as physical validation tools, verifying alignment with known electrochemical degradation mechanisms. Accepted at *Energy and AI*.

SyntheticPersonas Platform – Production AI System

2025

Architected a six-service production AI platform (FastAPI, React, PostgreSQL, MongoDB, Redis) deployed on Azure, generating synthetic survey respondents using LLM agent pipelines (LangChain, LangGraph, OpenAI/Anthropic APIs). Built complex synthetic data generation pipelines processing $\sim 11,000$ respondent profiles across ~ 338 variables, with evaluation harnesses validating outputs against ground-truth statistical distributions. Full CI/CD with GitHub Actions and Docker.

MPP-Inspector – Open-Source Developer Tooling for Stripe & Tempo

2025

Open-source contributor on the Machine Payments Protocol (co-authored by Stripe and Tempo) for AI agents. Published a TypeScript CLI toolkit and MCP server for inspecting, testing, and debugging MPP flows. Monorepo architecture with 156 unit and integration tests, mock and live endpoint test suites, CI via GitHub Actions, and direct integration with Claude Code and Cursor. [\[GitHub\]](#) [\[npm\]](#)

Research Publications

Journal Articles

Speeding Up MACE: Low-Precision Tricks for Equivariant Force Fields, **A. Benoit**, *arXiv preprint* (CS.ML), **2025** – [Arxiv](#)

Semi-Supervised Learning with Physicochemical Health Indicators Using an Encoder–Gaussian Process Framework for Battery State of Health Estimation under Dynamic Discharge Conditions, **A. Benoit**, P. Asef, **2025** – *accepted in Energy and AI*

Structural Invariance Matters: Rethinking Graph Rewiring through Graph Metrics, **A. Benoit**, C. Aitken, Y. He, *arXiv preprint* (CS.ML), **2025** – [Arxiv](#)

A Hybrid Artificial Intelligence Method for Estimating Flicker in Power Systems, J. Enayati, P. Asef, **A. Benoit**, **2025** – *Energy and AI*

Role of Structural Invariance in GNNs and Graph Transformers, C. Aitken, **A. Benoit**, H. Yu, **2025** – *incoming article in Learning on Graphs Conference (LoG)*

Navigating Intelligence: A Survey of Google OR-Tools and Machine Learning for Global Path Planning in Autonomous Vehicles, **A. Benoit**, P. Asef, **2024** – *Advanced Intelligent Systems*, vol. n/a, p. 2300840

Conference Papers

Intelligent Energy Management Using Multi-Agent Dynamic Learning for Scheduling Commercial Electric Vehicle Charging Stations, **A. Benoit**, K. Chan, P. Asef, **2024** – *59th International Universities Power Engineering Conference (UPEC)*

A Novel AI-Driven Hybrid Method for Flicker Estimation in Power Systems, J. Enayati, P. Asef, A. Yousefi, M. B. A., **A. Benoit**, **2024** – *International Conference on Smart Energy Systems and Technologies (SEST)*

Technical Skills

Programming Languages: Python • C/C++ • TypeScript • JavaScript • MATLAB • SQL

ML Frameworks: PyTorch • JAX (familiar) • TensorFlow • NumPy • SciPy • pandas • scikit-learn • GPyTorch

ML Systems & Infrastructure: GPU-accelerated training (NVIDIA RTX 3080 Ti clusters) • mixed-precision training (FP16/BF16/FP32) • cuEquivariance & fused equivariant kernels • Bayesian hyperparameter optimisation (Weights & Biases) • large-scale experiment tracking • systematic model benchmarking • synthetic data pipelines

LLM & Agentic Systems: LangChain • LangGraph • Langfuse • LangSmith • OpenAI API • Anthropic API • MCP • RAG • LLM Benchmarking • Multi-agent Workflows

DevOps & Infrastructure: Docker • Docker Compose • CI/CD (GitHub Actions) • Azure • Git • GitHub • FastAPI

Languages: French (Native), English (Fluent), Spanish (C1)